

Di Chai

Ph.D. Student of HKUST

Email: dchai@connect.ust.hk

[Homepage](#)

[Google Scholar](#)

Education

- Ph.D.** Hong Kong University of Science and Technology **Hong Kong, China** Sep 2019 - Jan 2025
Major: Computer Science and Engineering
Advisors: Prof. Kai Chen and Prof. Qiang Yang
- M.Sc.** Hong Kong University of Science and Technology **Hong Kong, China** Sep 2017 - Sep 2018
Major: Big Data Technology
- B.Sc.** University of Science and Technology Beijing **Beijing, China** Sep 2013 - Aug 2017
Major: Intelligent Science and Technology

Working Experience

- Research Assistant** Clustar & HKUST **Hong Kong, China** Sep 2018 - Aug 2019

Profile

Di Chai is a Ph.D. student at the Hong Kong University of Science and Technology, supervised by Prof. Kai Chen and Prof. Qiang Yang. His research interests focus on intelligent, secure, and high-performance distributed computation systems. Throughout his master's and Ph.D. studies, he has published six international conference and journal papers as the first author, including two papers with over three hundred citations. During his master's program, he concentrated on a spatio-temporal traffic prediction system in urban computing, contributing to one of the first systems that integrate graph neural networks with spatio-temporal traffic prediction. This work was presented at the prestigious GIS conference SIGSPATIAL'18 and achieved the highest citation count among all papers at that conference that year (over 300 citations). In his Ph.D. research, he has focused on large-scale distributed matrix factorization systems, which are foundational technologies for various real-world applications. His work was among the first to analyze privacy leakage issues in federated matrix factorization, leading to the proposal of a secure matrix factorization solution published in IEEE Intelligent Systems (over 300 citations). Subsequent research includes a federated singular value decomposition (SVD) system designed for billion-scale data, published in KDD'22, in collaboration with BGI Genomics. This system enables privacy-preserving whole-genome association analysis among multiple institutions, achieving lossless accuracy while maintaining high efficiency. Additionally, he has developed a decentralized efficient federated SVD system, accepted by ATC'24, which provides a high-performance federated matrix factorization solution without relying on external servers, marking the first secure SVD system that simultaneously achieves decentralization and high efficiency.

Undergoing Project

P1. Accelerate LLM Training through Token Filtering. (*submitted*)

Token filtering aims to enhance the utility of large language models (LLMs) by removing inconsequential tokens during training. However, previous methods have not achieved significant improvements due to limited sparsity from filtering only in output layers and inefficient sparse GEMM processes. We address these issues by filtering tokens across all layers and optimizing GEMM operations, resulting in notable reductions in backpropagation time (up to 35%) and overall training time (up to 22%). Evaluations show that our system improves model utility by 16% compared to standard training while reducing training time significantly, and it can be easily integrated into existing frameworks.

Conference Paper

C1. Efficient Decentralized Federated Singular Vector Decomposition.

Di Chai, Junxue Zhang, Liu Yang, Yilun Jin, Leye Wang, Kai Chen, and Qiang Yang.
USENIX ATC'24.

C2. Practical Lossless Federated Singular Vector Decomposition Over Billion-Scale Data.

Di Chai, Leye Wang, Junxue Zhang, Liu Yang, Shuowei Cai, Kai Chen, and Qiang Yang.
ACM SIGKDD'22.

C3. Bike Flow Prediction with Multi-Graph Convolutional Networks.

Di Chai, Leye Wang, and Qiang Yang.
SIGSPATIAL/GIS'18. [Google Scholar 300+ Citations]

C4. Sphinx: Enabling Privacy-preserving Online Learning over the Cloud.

Han Tian, Chaoliang Zeng, Zhenghang Ren, Di Chai, Junxue Zhang, Kai Chen, and Qiang Yang.
IEEE S&P'22.

Journal Paper

J1. A Survey for Federated Learning Evaluations: Goals and Measures.

Di Chai*, Leye Wang*, Liu Yang, Junxue Zhang, Kai Chen, and Qiang Yang. (*Co-first Authors)
IEEE TKDE Accepted (2024).

J2. Secure Federated Matrix Factorization.

Di Chai, Leye Wang, Kai Chen, and Qiang Yang.
IEEE Intelligent Systems, 36(5): 11-20 (2021). [Google Scholar 300+ Citations]

J3. Efficient Federated Matrix Factorization against Inference Attacks.

Di Chai, Leye Wang, Kai Chen, and Qiang Yang.
ACM TIST, 2022, 13(4): 1-20.

J4. Exploring the Generalizability of Spatio-Temporal Traffic Prediction: Meta-Modeling and an Analytic Framework.

Leye Wang, Di Chai, Xuanzhe Liu, Liyue Chen, and Kai Chen.
IEEE TKDE, 2021, 35(4): 3870-3884.

Workshop Paper

W1. Aegis: A Trusted, Automatic and Accurate Verification Framework for Vertical Federated Learning.

Cengguang Zhang, Junxue Zhang, Di Chai, and Kai Chen.
IJCAI FL-Workshop (2021). [Best Application Award]

W2. Practical and Secure Federated Recommendation with Personalized Mask.

Liu Yang, Junxue Zhang, Di Chai, Leye Wang, Kun Guo, Kai Chen, and Qiang Yang.
International Workshop on Trustworthy Federated Learning (2022).

W3. Secure Forward Aggregation for Vertical Federated Neural Networks.

Shuowei Cai, Di Chai, Liu Yang, Junxue Zhang, Yilun Jin, Leye Wang, Kun Guo, and Kai Chen.
International Workshop on Trustworthy Federated Learning (2022).

Honors and Awards

2022 Best Application Awarers @ International Workshop on Trustworthy Federated Learning

2017 SODA Seed Award (with ten thousand yuan bonus)

2016 Meritorious Winners, Mathematical Contest In Modeling (Top 8%)

Journal Reviews

I have been a reviewer for the following journals:

- IEEE Transactions on Mobile Computing.
- IEEE Transactions on Services Computing.
- IEEE Transactions on Information Forensics and Security